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Artificial Intelligence-Driven Development of Nickel-Catalyzed Enantioselective Cross-Coupling Reactions

Yadong Gao,[#] Kunjun Hu,[#] Jianhang Rao, Qiang Zhu, and Kuangbiao Liao*

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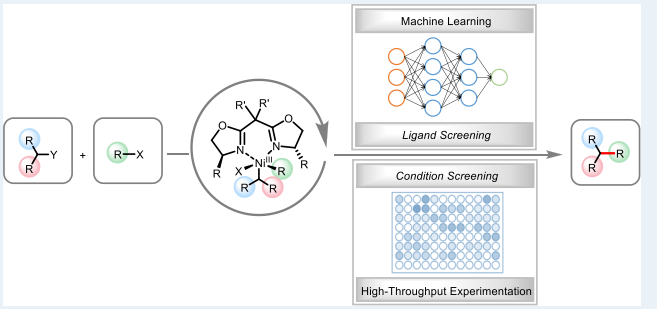
ABSTRACT: The conventional approach to developing asymmetric synthetic methods relies heavily on empirical optimization. However, the integration of artificial intelligence (AI) and high-throughput experimentation (HTE) technology presents a paradigm shift with immense potential to revolutionize the discovery and optimization of asymmetric reactions. In this study, we present an efficient workflow for the development of a series of nickel-catalyzed asymmetric cross-coupling reactions, leveraging AI and HTE technology. Many nickel-catalyzed enantioselective cross-coupling reactions share a common Ni(III) intermediate, which dictates the enantioselectivity. To harness this mechanistic insight, we embarked on developing a predictive model for nickel-catalyzed enantioselective coupling reactions, elucidating the general rules governing enantioselectivity. Through the application of data science tools and HTE technology, we curated a data set to construct an AI-based model. This model was subsequently utilized to facilitate the discovery of efficient nickel hydride-catalyzed enantioselective and regioselective cross-coupling reactions. Employing AI-assisted virtual ligand screening and HTE-enabled condition optimization, we successfully identified optimal ligands for eight coupling reactions. Consequently, a series of chiral sp^3 C−C bonds were synthesized with high yield and enantioselectivity.



KEYWORDS: *artificial intelligence, machine learning, high-throughput experimentation, nickel, enantioselective, cross-coupling, AutoGluon*

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■ INTRODUCTION

The importance of asymmetric reactions has been well recognized by the 2001 and 2021 Nobel Prizes in Chemistry.[1] A chiral catalyst or ligand plays a key role in controlling asymmetric reactions. However, identifying an optimal chiral catalyst or ligand for a specific transformation has been slow because it has generally not been a rational design process, relying mainly on chemical intuition, empiricism, and screening.[2] In recent years, there has been a surge of research focused on computational science approaches to chiral catalyst or ligand design.[3] When integrated with traditional experimental approaches, the use of cheminformatics, artificial intelligence (AI), and automation techniques shows great potential for understanding new enantioselective reactions and revolutionizing the process of developing new asymmetric reactions.[4]

In addition to the catalyst or ligand, a variety of reaction parameters such as solvent, additive, temperature, substrate interactions, conformation, and steric or electronic considerations can significantly affect the reactivity and selectivity of an enantioselective reaction.[2b] Reaction optimization is an inevitable but time-consuming and labor-intensive process in the development and application of synthetic methods. Fortunately, over the past decade, high-throughput experimentation (HTE) has emerged as a powerful tool for organic synthesis in industry and academia.[5] HTE allows for the rapid execution of large numbers of experiments for screening and accelerates the collection of scaled-up data for AI.[6] Integrating HTE for large-scale experiment data and AI for prediction of yield or enantiomeric excess (ee) value into a workflow is an increasingly popular strategy that takes advantage of the complementary nature of these tools and provides numerous opportunities to explore chemical space.[7]

Naturally abundant sp^3-carbon stereocenters are widespread in diverse natural products, pharmaceuticals, and materials.[8] Therefore, the development of methods that can simultaneously construct sp^3-carbon centers and control stereochemistry is crucial, albeit challenging. Over the past two decades, significant progress has been made in the field of transition metal-catalyzed enantioselective cross-coupling reactions,[9] with the nickel catalyst in particular attracting considerable attention due to its unique catalytic activity.[10]

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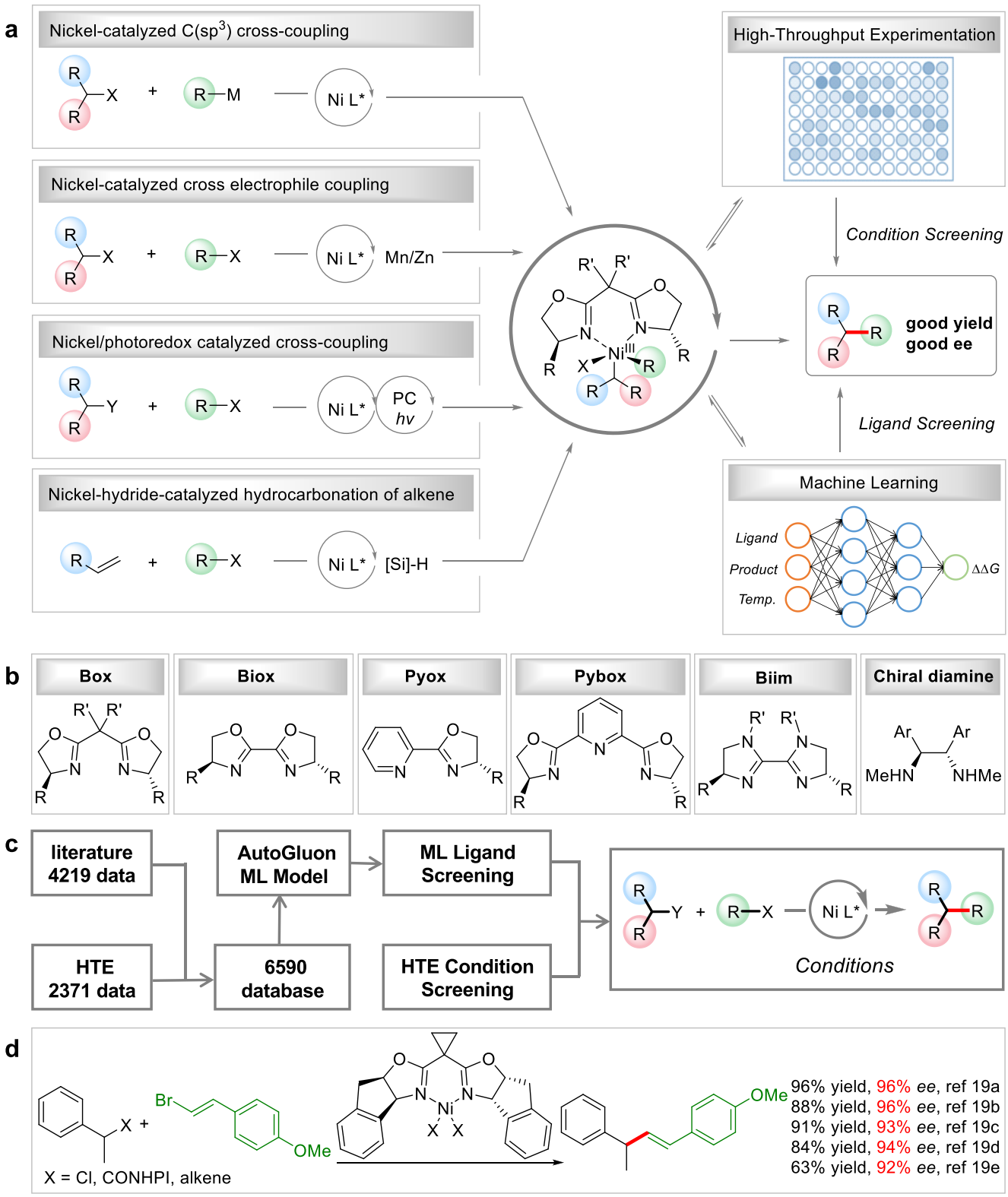

Figure 1. (a) Nickel-catalyzed enantioselective cross-coupling reactions; (b) chiral nitrogen ligands; (c) workflow for the development of nickel-catalyzed enantioselective cross-couplings; (d) literature data on different methods for producing the same product.

Since the pioneering work of Fu and co-workers,[11] elegant methods have been developed for the construction of chiral C(sp^3)-hybridized centers, such as nickel-catalyzed enantioselective cross-couplings,[12] cross-electrophilic couplings,[13] nickel/photoredox-catalyzed enantioselective couplings,[14] and nickel hydride-catalyzed enantioselective hydrocarbonation of alkenes[15] (Figure 1a).

In light of the mechanism underlying several nickel-catalyzed enantioselective cross-coupling reactions, it is generally agreed that these transformations typically proceed through a Ni(III) intermediate prior to the reductive elimination of the coupling

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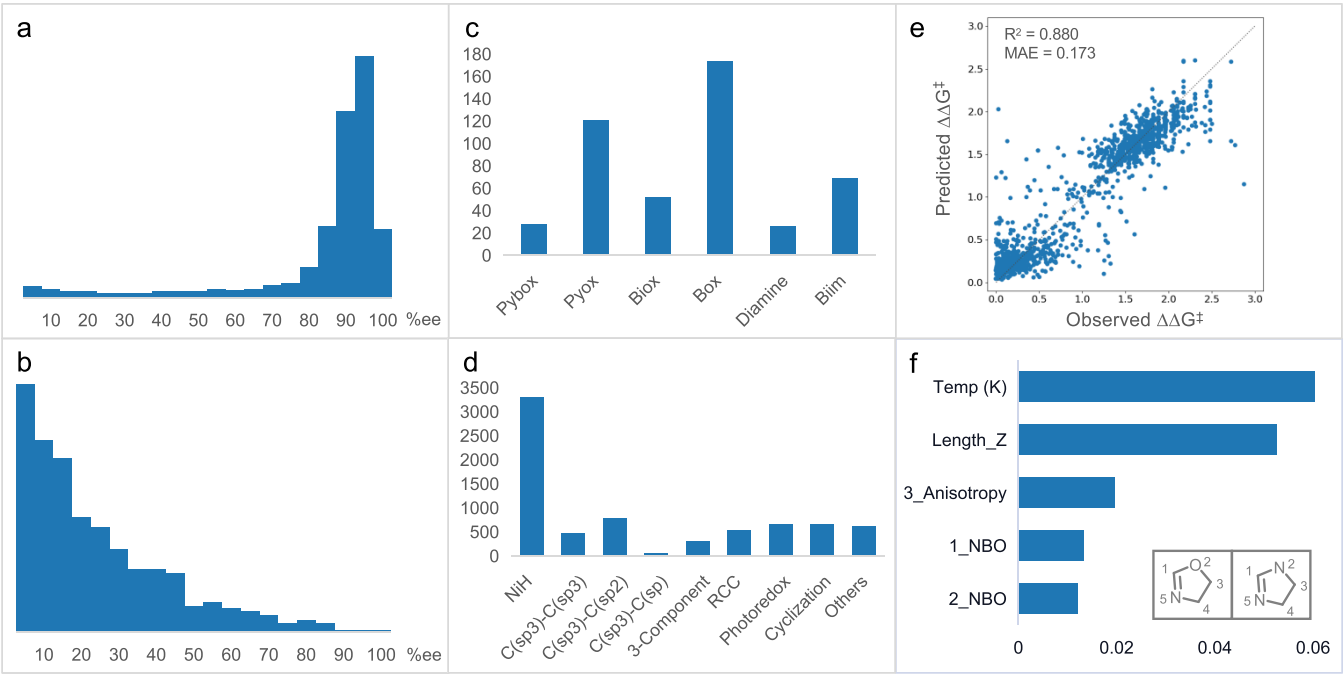

Figure 2. (a) Distribution of literature data; (b) distribution of HTE data; (c) distribution quantity of ligand type of the entire data set; (d) distribution quantity of reaction type of the entire data set; (e) observed versus predicted plots for the AutoGluon model and linear regression analysis; (f) top 5 ranking of the feature importance scores in the AutoGluon model with DFT descriptors.

products[10a,b,13c,14c,15a,16] (Figure 1a). The key intermediate responsible for determining the enantioselectivity of the coupling products is the Ni(III) complex bearing chiral ligands and carbon groups. This Ni(III) complex creates a specific chiral environment with both chiral ligands and substrate fragments. Unlike palladium catalysts, which typically use phosphorus ligands,[17] nickel catalysts more commonly employ nitrogen ligands such as oxazoline ligands (e.g., Box, Biox, Pyox, and Pybox), imidazoline ligands, and chiraldiamine ligands[18] (Figure 1b). It is likely that there is a general rule governing the enantioselectivity of nickel-catalyzed coupling reactions that could be elucidated by AI and help chemists develop novel asymmetric reactions.

Herein, we present an efficient workflow for the development of nickel-catalyzed enantioselective cross-couplings by combining AI for ligand screening and HTE for condition screening. A series of asymmetric sp^3 C−C bonds were formed via a nickel-catalyzed enantio- and regioselective transformation through an effective HTE screening guided by the AI-based model and mechanism considerations (Figure 1c).

■ RESULTS AND DISCUSSION

To initiate this workflow, we curated an extended inventory of 4219 nickel-catalyzed enantioselective reactions from 117 published literature before 2023 (Table S1). Upon thorough analysis, different methods were able to obtain the same product; interestingly, almost identical enantiomeric excess values were obtained when using the same ligand, although the substrates and conditions were quite different (Figure 1d and Figure S1).[19] Based on these preliminary data, we hypothesized that connecting common Ni(III) mechanistic features through simultaneous interrogation of ligands and products would provide a holistic view of the key interactions responsible for the observed enantioselectivity of these reactions. We therefore decided to develop an AI-based model by using only the descriptors related to the chiral ligands and the coupling products as inputs without taking into account reaction parameters such as substrates, catalysts, solvents, and additives. This model can unveil the relationship between ligands and products in nickel-catalyzed cross-couplings for the early virtual screening of chiral ligands.

It is worth mentioning that the use of a Python script to extract information on the International Union of Pure and Applied Chemistry (IUPAC) name and enantiomeric excess (% ee) of products from the Supporting Information in the related literature makes it easy to obtain most of the data required for AI-based model development. By transferring IUPAC names into the Simplified Molecular Input Line Entry System (SMILES)[20] format, molecular structures can be recognized by RDKit.[21] This approach significantly streamlines data extraction by eliminating the need for complex collection and processing of substrate and condition parameter. With a total of 4219 literature data collected, various machine learning (ML) algorithms were applied to identify correlations between the molecular structure and reported enantiomeric excess. Using the molecular fingerprints of ligands and products and the reaction temperature as inputs and the transition state free energy $\Delta\Delta G^{\ddagger}$ as output ($\Delta\Delta G^{\ddagger} = -RT \ln(\mathrm{er})$, where er denotes the enantiomeric ratio, T represents the reaction temperature, and R is the gas constant), several machine learning algorithms were used to establish predictive models for nickel-catalyzed enantioselective cross-couplings. However, unsatisfactory results were obtained after the initial exploration (Table S5), which might be due to the highly unbalanced distribution of good results in the literature data, which might be due to literature bias in reporting positive results only and overlooking negative data points (Figure 2a). Therefore, a more comprehensive data set was considered necessary for further ML analysis.

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HTE has emerged as a powerful tool for building an experimental data set by enabling the rapid execution of a large number of experiments.[5,6] To solve the problem of insufficient literature data, 72 sets of high-throughput experiments from the relevant literature were performed at the 10 μmol scale in 96-well plates using the HTE workflow (Figure S2).[22] Each substrate set was reacted with 47 commercially available chiral nitrogen ligands (Figure S8) under optimized literature conditions, resulting in 3384 completed reactions using automated instrumentation. We used Chemspeed for the loading of catalysts and additives, including solid additives, and the TECAN robotic liquid handling workstation for the transfer of prepared substrate solutions and solvents. The yields of all of the reactions were analyzed using the ultraperformance liquid chromatography mass spectrometry (UPLC-MS) high-throughput analytical system, while the enantioselectivities of the reactions producing the target products were analyzed using the ultraperformance convergence chromatography mass spectrometry (UPCC-MS) high-throughput analytical system. Finally, we obtained 2371 enantiomeric excess experiment data for nickel-catalyzed enantioselective cross-couplings after the high-throughput analytical process, since not all reactions yield the corresponding coupling products (Figure S3). Further analysis showed that the HTE data contained a large number of negative results (Figure 2b), and we speculate that HTE data could serve as an effective complement to the literature data.

With the fusion data set in hand, ML-based prediction models were used to determine the effect of ligands on specific products. Figure 2c,d shows the distribution of ligand and reaction types within the full data set of 6590 entries, comprising 380 ligands and 3385 products with different structures. While computational modeling,[23] three-dimensional (3D) quantitative structure−activity relationship,[2b] and multivariate linear regression[24] offer valuable assistance in expediting the exploration of asymmetric catalysis and ligands, the computation of quantitative descriptors remains a time-consuming process, limiting the practical applicability of such models. This challenge becomes even more pronounced for complex, large data-based models. Therefore, we decided to develop a practical prediction model that uses straightforward cheminformatics processing including a molecular fingerprint representation. Incorporating with the molecular fingerprints transmitted by molecular additive fingerprints (MAF),[22] a molecular fingerprint representation developed by our research group, Gradient Boosted Trees (GBT), Extreme Gradient Boosting (XGB), and Random Forest Regression (RF) algorithms were applied using KNIME,[25] an open-source data analytics platform (Figure S13). A random split of the entire data set and a 5-fold cross-validation procedure were conducted to evaluate the ML-based models. The combination of MAF based on 1024 size and RF algorithms was found to give the best performance, with an R^2 of 0.834 and a mean absolute error (MAE) in $\Delta\Delta G^{\ddagger}$ of 0.215 kcal/mol (Table 1, entry 3).

AutoGluon,[26] developed by Amazon Ltd., is an open-source auto-ML toolkit that is both easy to use and easy to extend. It has the ability to automatically select algorithms, train models, and optimize hyperparameters with high efficiency. The AutoGluon Tabular Predictor function was used to construct the prediction model using the full data set. Fortunately, a significantly better model was obtained with an R^2 value of 0.844 and an MAE in $\Delta\Delta G^{\ddagger}$ of 0.199 kcal/mol (Table 1, entry 4). This may be due to the fact that AutoGluon uses a multilayer stack ensemble strategy with a series of customized models, which can provide better results than any single algorithm.

Table 1. Performance of the ML Models

entry	algorithm	descriptor	R^2	MAE (kcal/mol)
1	GBT	MAF	0.803	0.233
2	XGB	MAF	0.831	0.213
3	RF	MAF	0.834	0.215
4	AutoGluon	MAF	0.844	0.199
5	AutoGluon	RDKit	0.830	0.214
6	AutoGluon	DFT	0.806	0.233
7	AutoGluon	MAF, RDKit, and DFT	0.842	0.203
8	GraphRXN	Graph	0.769	0.259
9[a]	AutoGluon	MAF	0.880	0.173

[a]The model was established using 6380 data points with 170 ligands by excluding the data related to 210 ligands that only appeared once.

Considering that the current model may not be sufficient for practical applications, more cheminformatics processing methods were used to verify the effectiveness of the model. 2D descriptors calculated using RDKit, as well as molecular and atomic descriptors calculated using density functional theory (DFT) calculations, were applied to generate the AutoGluon model (see the Supporting Information for details of the RDKit and DFT descriptors). Unfortunately, no better results were obtained, although considerable computational resources were consumed (Table 1, entries 5 and 6). Even when the calculated descriptors were combined with the MAF and RDKit descriptors, a worse result was obtained (Table 1, entry 7). In addition, the graph model network GraphRXN,[27] which is based on a deep learning algorithm, was used but still failed to give a better performance (Table 1, entry 8).

It is worth noting that a ring in imidazoline ligands and oxazoline ligands was numbered during the DFT atomic descriptor calculation (Figure S12). Also, an interpretable ML model based on DFT descriptors was utilized to gain a better understanding of the AutoGluon model (Table S19). The top 5 ranking analysis of the feature importance scores in the model (Figure 2f) revealed that the reaction temperature was the most important factor, which is a reasonable finding as the impact of temperature on enantioselectivity is quite evident, and the impact is not consistent for different reactions.[28] Moreover, the length of the ligands and the anisotropy of atom 3, as well as the NBO charge of atoms 1 and 2 of the ligands, all carried high importance in the prediction model, indicating the crucial role of the ligand structures in the coupling reactions. Due to the type of imidazoline and oxazoline, ligands were directly distinguished by atom 2, and atoms 1 and 3 were vicinal to atom 2. We believe that the model could indeed identify the structure of the ligand molecules and provide a reasonable prediction.

We have also endeavored to segment the data set using a 70:30 ratio across various strategies, encompassing multiple random splits, the Kennard Stone algorithm, the *y*-equidistant technique, linear sampling, and stratified sampling. Upon rigorous evaluation of kinds of algorithms and descriptors, our findings consistently highlight that leveraging the MAF descriptor in conjunction with AutoGluon modeling yields the most optimal outcomes (Table S20). Further in-depth optimization led to the best model only using 6380 data points

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with 170 ligands by excluding the data related to 210 ligands that only appeared once. The resulting AutoGluon model achieved an R^2 of 0.880 and an MAE in $\Delta\Delta G^{\ddagger}$ of 0.173 (Table 1, entry 9, and Figure 2e). Other machine learning algorithms can generally also provide better results for the 6380 data points (Table S11). Those ligands that only appeared once were not widely used; therefore, we believe that the optimized model may be more conducive to practical application.

With the best-performing AutoGluon model based on MAF descriptors in hand, the data of recently reported nickel-catalyzed enantioselective cross-coupling reactions since 2023,[29] which are a practical external verification set, have been used to verify the performance of the AutoGluon model. We wanted to know whether the model could predict the optimal ligands for these new reactions. The predicted results of chiral ligands of the representative coupling products in the literature were compared with the experimental results in the literature. As shown in Table 2, the results were quite satisfactory (see Table S23 for details). The absolute error between the experimental ee values in the literature and the predicted ee values of the model is mostly within 10%, and even a few errors are within 5%. Moreover, the optimized ligands in this literature are mostly in the top 10 rank in the library of 170 ligands. That is to say, machine learning models could be used to assist in chiral ligand screening in the ligand optimization process, and only a very small number of ligands need to be attempted before obtaining the optimized ligands. Based on these results, we believe that the established prediction model holds the potential for chiral ligand screening in the development of new nickel-catalyzed enantioselective synthesis. Therefore, we attempted to apply this ML model in the discovery and optimization of new nickel-catalyzed enantioselective reactions.

Table 2. Performance of the AutoGluon ML Model on the Literature Data

lit.	lit. ligand	lit. temp. (K)	lit. data (% ee)	pred. data (% ee)	pred. error	pred. rank
29a	**L312**	298	96	94	2	1
29b	**L437**	298	99	87	12	2
29c	**L437**	263	97	94	3	3
29d	**L162**	298	92	94	2	4
29e	**L162**	273	92	91	1	7
29f	**L437**	298	94	85	9	10
29g	**L210**	268	92	89	3	14
29h	**L162**	313	93	85	8	15
29i	**L162**	298	90	87	3	24
29j	**L162**	298	94	85	9	28

The optimized AutoGluon model was first applied to reactions reported in the literature. A table of target products and 170 ligands processed by MAF was generated, and the prediction model was used to determine the corresponding $\Delta\Delta G^{\ddagger}$ of the products with each ligand. The model-predicted well-performing ligand **L162** was used in the reactions for the target products, leading to the successful synthesis of the chiral product **P1** with an 86% yield and 71% ee, by replacing the achiral phenanthroline ligand **L01** (Figure 3a).[30] Although there was a slight discrepancy between the experimental and predicted results, this strategy demonstrated the advanced capabilities of the ML model in guiding the quick development of an asymmetric synthesis. Furthermore, the model-predicted well-performing **L163** was used for a literature-reported asymmetric synthesis, giving the chiral product **P2** with a comparable 75% yield and better 87% ee, despite the significant difference between the imidazoline ligand **L163** and the oxazoline ligand (**L137**) in the literature.[31] The predicted value in this case showed excellent agreement with the experimental value (Figure 3b). This approach directly provided an alternative highly efficient ligand for a specific enantioselective reaction, circumventing the need for large-scale screening in the traditional optimization process.

Building on the positive results obtained in the previous experiments, the focus of the study then shifted toward developing new Ni-catalyzed enantioselective cross-coupling reactions to demonstrate the potential of AI and HTE. Chiral amines and related amide derivatives are privileged motifs in many natural products and pharmacologically active molecules.[32] Although there are already elegant synthetic methods for chiral amine compounds,[33] developing methods with broad applicability will greatly expand the accessibility of such compounds. Therefore, we attempted to develop a general nickel-catalyzed enantioselective cross-coupling synthetic method for chiral amines based on AI and HTE technologies. In recent years, there have been significant advances in the NiH-catalyzed reductive hydrocarbonation of alkenes, enabling both enantioselective and remote hydrofunctionalization to be achieved in a regio- and enantioselective manner.[15] Two enamines (**SM1** and **SM2**) were selected to react with a range of organic electrophiles, including aryl, alkenyl, alkyl, benzyl, and allyl halides, by using the NiH-catalyzed reductive hydrocarbon method. HTE was employed to investigate these racemic reactions under the condition of a preprepared nickel catalyst, silane, KF, and dimethylacetamide solvent at room temperature. Fortunately, most of the corresponding coupling products could be detected by UPLC-MS (Figure 3c), which fully illustrates the universality of the NiH-catalyzed reductive hydrocarbon method.

Eight well-performing reactions were arbitrarily selected for the development of NiH-catalyzed enantioselective cross-couplings. Using the prediction process, the optimized AutoGluon model based on MAF descriptors produced an enantioselective relationship between 170 ligands and the corresponding products. Figure 3d displays the top 5 best-performing ligands for the eight products (**P3**−**P10**). Experimental validation of the reactions leading to the formation of these eight products was conducted using the top 5 ligands predicted by AI. The results revealed that the majority of data points exhibited ee values greater than 50%, with the best result exceeding 90% (Figure 3e, red). In contrast, when 47 randomly selected chiral ligands were employed to evaluate the enantioselectivity of these reactions through HTE, only a few results showed ee values below 50%, and most of them showed no selectivity (Figure 3e, blue). In addition to this, we also tested the chiral ligands that appeared most frequently in the data set, as well as those with the highest average ee values (Figures S9−S11), which are typically prioritized based on chemical intuition. However, the performance of these ligands in these reactions was not satisfactory except **L437** and **L162**; these two ligands are also among the AI top 5 ligands (Figure 3e, green and yellow). These experimental validation data clearly demonstrate that AI-predicted ligands significantly outperform those selected by human experience (see Figure S25 for details). These results serve as evidence of the effectiveness of the ML model,

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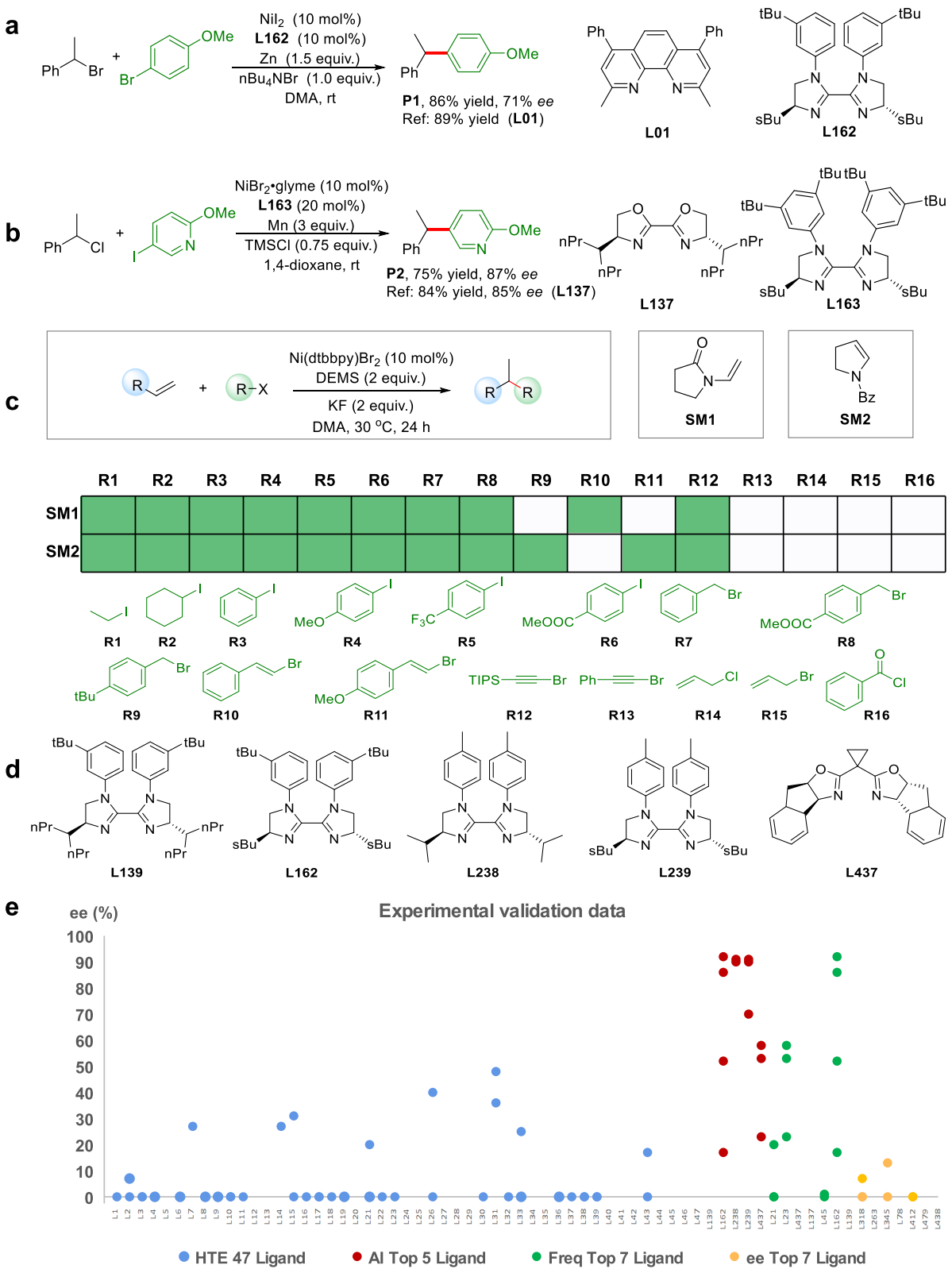

Figure 3. Data for nickel-catalyzed enantioselective cross-coupling reactions. (a) Synthesis of chiral products **P1**; (b) synthesis of chiral products **P2**; (c) discovery of new Ni-catalyzed enantioselective cross-coupling reactions. The green squares represent successful reactions, while the white squares represent unsuccessful reactions. (d) top 5 AI ligands; (e) experimental validation data of HTE 47 ligands, AI top 5 ligands, frequency top 7 ligands, and average ee value top 7 ligands. DMA, dimethylacetamide; TMSCl, trimethylsilyl chloride; DEMS, diethoxymethylsilane. See the Supporting Information for details.

enabling efficient virtual screening of chiral ligands and minimizing the need for extensive random screening efforts.

A 24-well HTE sequence with 4 nickel catalysts and 6 silanes was designed to screen reaction conditions using the

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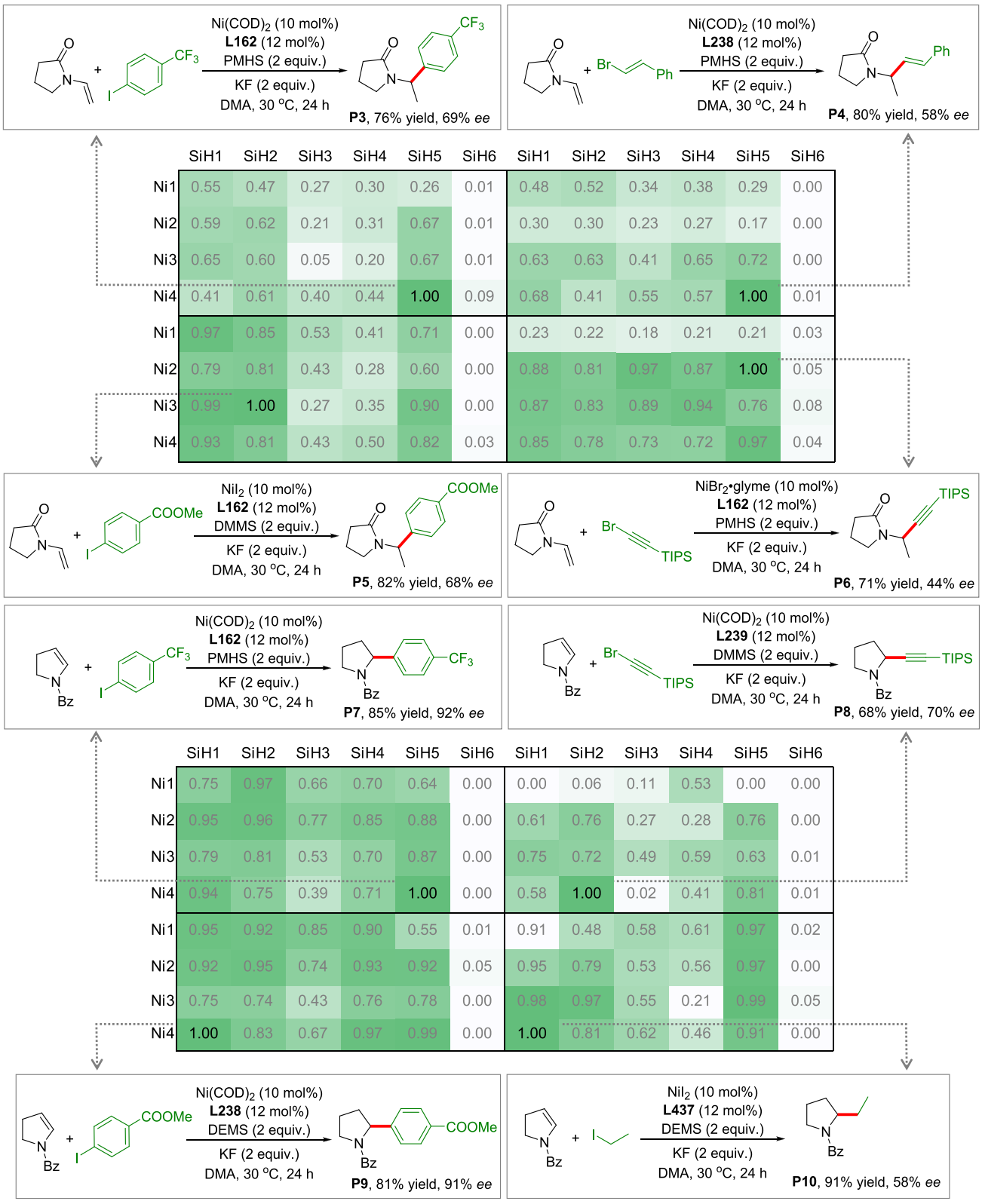

Figure 4. Condition screening for the generation of products **P3**−**P10**. The heat map represents the ratios of products to the internal standard; the maximum ratio was set as 1.00. Internal standard, ethyl-1,1′-biphenyl; **Ni1**, $NiCl_2$·glyme; **Ni2**, $NiBr_2$·glyme; **Ni3**, NiI_2; **Ni4**, $Ni(COD)_2$; **SiH1**, DEMS; **SiH2**, DMMS (dimethoxymethylsilane); **SiH3**, $(EtO)_3SiH$ (triethoxysilane); **SiH4**, $(MeO)_3SiH$ (trimethoxysilane); **SiH5**, PMHS (poly(methylhydrosiloxane)); **SiH6**, Et_3SiH (triethylsilane). See the Supporting Information for details of the reactions.

corresponding best chiral ligand for the starting material **SM1**. The relative ratios of the desired products to the internal standard detected by UPLC-MS are listed in Figure 4. The optimized reaction conditions with the corresponding nickel catalyst and silane were employed in 0.2 mmol scale reactions, yielding the coupling products with high efficiency and

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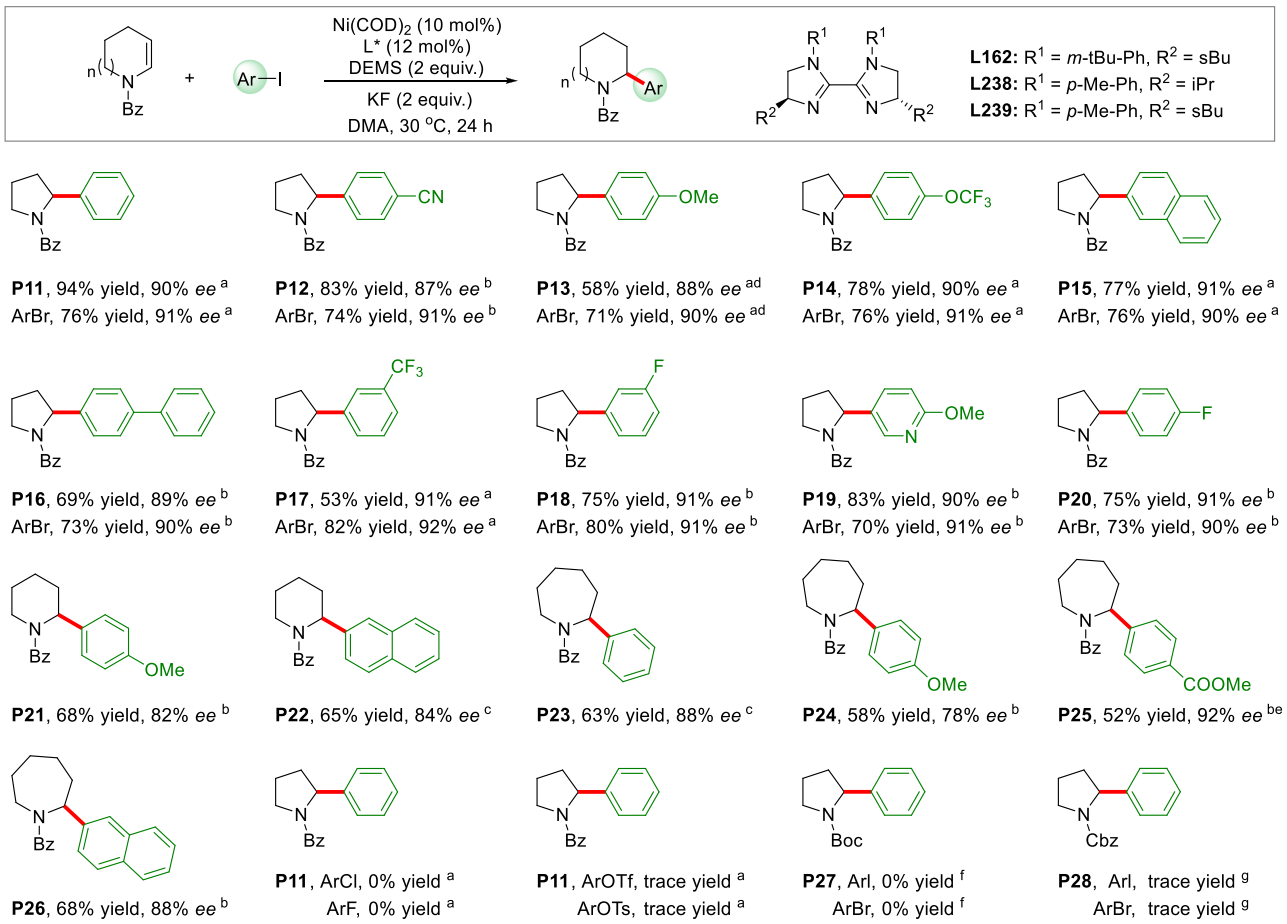

Figure 5. Substrate scope for nickel-catalyzed enantioselective arylation. (a) **L238** as a ligand; (b) **L162** as a ligand; (c) **L239** as a ligand; (d) PMHS as silane; (e) DMMS as silane; (f) AI top 5 ligands for **P27** were used; (g) AI top 5 ligands for **P28** were used. See the Supporting Information for details of the reactions.

moderate enantiomeric excesses. Specifically, we gave arylated **P3** with a 76% yield and 69% ee using **L162**, alkenylated **P4** with an 80% yield and 58% ee using **L238**, arylated **P5** with an 82% yield and 68% ee using **L162**, and alkynylated **P6** with a 71% yield and 44% ee using **L162**. Notably, although **SM1** reacted with aryl iodides **R5** and **R6**, different optimal conditions were obtained, suggesting that HTE can tailor the conditions for a specific target product. Similarly, the desired coupling products were synthesized under the optimized reaction conditions for **SM2**, yielding arylated **P7** with an 85% yield and 92% ee using **L162**, alkynylated **P8** with a 68% yield and 70% ee using **L239**, arylated **P9** with an 81% yield and 90% ee using **L238**, and alkylated **P10** with a 91% yield and 58% ee using **L437** (Figure 4).

The AI- and HTE-assisted development of the nickel-catalyzed enantioselective arylation method of **SM2** was impressive. Therefore, the scope of this reaction was investigated (Figure 5). By utilizing the AutoGluon model to predict the optimal ligands for specific products, followed by HTE for rapid screening of reaction conditions, we successfully obtained a series of chiral amine compounds with good yields and enantioselectivities. A series of aryl iodides with various functional groups, such as CN, OMe, OCF_3, CF_3, F, naphthyl, and biphenyl groups, were compatible with the reaction conditions and gave the corresponding chiral arylation products (**P11**−**P20**) with good yields and ee values. The effects of electron-withdrawing groups and electron-donating groups on the reaction are not obvious, indicating that the aryl iodides may undergo a nonrate-determining step (Figure S54). It is worth noting that substituted pyridine iodides can also smoothly obtain the corresponding arylation products (**P19**). In addition, the six-membered piperidine substrates and the seven-membered azepane substrates can also be compatible with this reaction system, obtaining a series of chiral arylation products (**P21**−**P26**) with different ring systems with moderate yields and good enantioselectivity, fully demonstrating the great application potential of this reaction strategy for the synthesis of chiral amine compounds. It is worth noting that the optimized chiral ligands (**L238**, **L162**, and **L239**) used in these enantioselective arylation reactions are all the top 5 ligands predicted by the ML model, and the optimal ligands and silane corresponding to different substrates are slightly different, which further demonstrates the application potential of AI and HTE in customizing the best reaction conditions for specific substrates.

It is interesting that aryl bromides, alongside aryl iodides, can effectively act as coupling partners in nickel-catalyzed enantioselective cross-coupling reactions, yielding the corresponding coupling products with comparable yields and ee values to those obtained with aryl iodides; even in some cases, better enantioselectivity can be achieved (**P11**−**P20**). In contrast, aryl chloride-, aryl fluoride-, and sulfonate-based pseudohalide substrates do not participate well in the coupling reactions. For alkene substrates, we have also attempted other

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types of protecting groups, such as Boc and Cbz, but unfortunately, these substrates did not yield satisfactory results, with almost no corresponding products obtained under standard reaction conditions using the top 5 ligands predicted by the AutoGluon model.

While HTE has proven highly efficient for condition screening, AI for ligand screening has not yielded similarly satisfactory results. This may be due to limitations in predictive accuracy of the AI model or to the fact that the set of 170 ligands used for testing may not provide an optimal outcome for these reactions. While 170 ligands represent a significant database, it still may not fully encompass the vast chemical space of potential chiral ligands. When the current library of 170 ligands fails to yield satisfactory results for specific reactions, it is worth considering the expansion of the ligand library to discover more effective ligands for the developed reactions.

The shortcomings of the current AI models may be due to the inability of current molecular fingerprints, RDKit descriptors, and DFT descriptors to accurately describe molecular structures, particularly for chiral molecules. For example, the @ symbol in SMILES has been used to denote the relative chirality of molecules, but a comparable AI model was found to be achievable, even when using SMILES without the @ symbol (Table S17). It has been demonstrated that racemic structures can also be used in the enantioselectivity prediction model. However, SMILES expressions and the corresponding molecular fingerprints derived from SMILES have a limited ability to accurately identify chiral structures. We propose that the development of more effective molecular descriptors, especially 3D molecular descriptors, is a promising area for further research in cheminformatics. Descriptors based on DFT calculations may be able to describe 3D molecular structures more accurately; however, it is important to recognize that while computational resources for constructing regression models with various algorithms are easily accessible for data sets containing thousands of data points, the calculation of DFT descriptors for a similar number of compounds requires significant computational power (Tables S21 and S22). As the size of data sets increases, the computational resources for DFT calculations become a critical consideration. In contrast, acquiring MAF or RDKit descriptors is significantly more straightforward. Consequently, we hold the view that the development of an efficient and precise descriptor calculation methodology is essential for advancing the application of AI in chemical research.

At last, we believe that the findings presented indicate that our approach to constructing a unified model based on the common Ni(III) mechanistic features in nickel-catalyzed enantioselective cross-coupling reactions is not only viable but also effective. By focusing exclusively on the essential chiral ligands and the coupling products, the model elegantly sidesteps the complexity of various secondary factors, including substrates, catalysts, solvents, and reductants, to correlate the chiral ligand with the product's enantioselectivity. This streamlined model is poised to facilitate predictions for existing reactions and guide the development of ligands for novel reactions. We are optimistic that this methodology could be extended to the development and refinement of a wider array of enantioselective reactions.

■ CONCLUSIONS

In summary, we present an efficient workflow combining HTE and AI for the development of novel nickel-catalyzed enantioselective cross-coupling reactions. An AutoGluon model based on MAF descriptors was used to guide efficient chiral ligand screening, while HTE was employed as a powerful tool for reaction parameter optimization. Based on the reaction mechanism, especially the key Ni(III) intermediate, a regression model for reaction enantioselectivity can be accurately established by using only the descriptors of chiral ligands, coupling products, and reaction temperature as inputs. The machine learning model not only accurately predicts the optimized ligand reported in the recent literature but also could be applied to the chiral ligand optimization of new nickel-catalyzed enantioselective cross-coupling reactions. A series of chiral sp^3 C−C bonds were generated with good yield and enantiomeric excesses using a nickel-catalyzed enantio- and regioselective cross-coupling method. We envision that this innovative workflow will serve as a model for enantioselective method development and inspire researchers to further explore the synthetic potential of HTE and AI in this field.

■ MATERIALS AND METHODS

All information pertaining to the materials and methods used in this study is provided in the Supporting Information.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c04277.

Detailed information on machine learning method, DFT calculations, experimental procedures, characterization data, and spectroscopic data (PDF)

Accession Codes

The accession codes are available free of charge at https://github.com/TheLiaoGroup/Enantioselective-Cross-Coupling-Prediction (Python script, KNIME workflow, raw data set, MAF descriptors, RDKit descriptors, DFT descriptors, and calculated coordinates).

■ AUTHOR INFORMATION

Corresponding Author

Kuangbiao Liao − *Guangzhou National Laboratory, Guangzhou 510005, China; Bioland Laboratory, Guangzhou 510005, China;* orcid.org/0000-0001-9089-0569; Email: liao_kuangbiao@gzlab.ac.cn

Authors

Yadong Gao − *Guangzhou National Laboratory, Guangzhou 510005, China; Bioland Laboratory, Guangzhou 510005, China; Institute of Materia Medica, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing 100050, China*

Kunjun Hu − *Guangzhou National Laboratory, Guangzhou 510005, China; MOE Laboratory of Bioinorganic and Synthetic Chemistry, GBRCE for Functional Molecular Engineering, LIFM, IGCME, School of Chemistry, Sun Yat-Sen University, Guangzhou 510275, China*

Jianhang Rao − *Guangzhou National Laboratory, Guangzhou 510005, China*

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Qiang Zhu − *Guangzhou Institutes of Biomedicine and Health, Guangzhou 510530, China;* orcid.org/0000-0002-1243-2391

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.4c04277

Author Contributions

[#]Y.G. and K.H. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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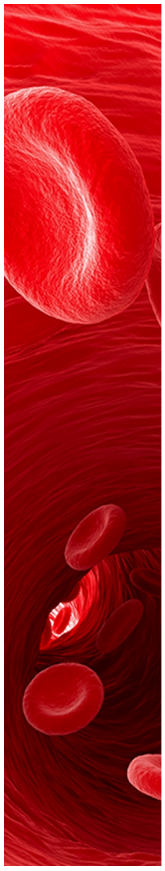

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